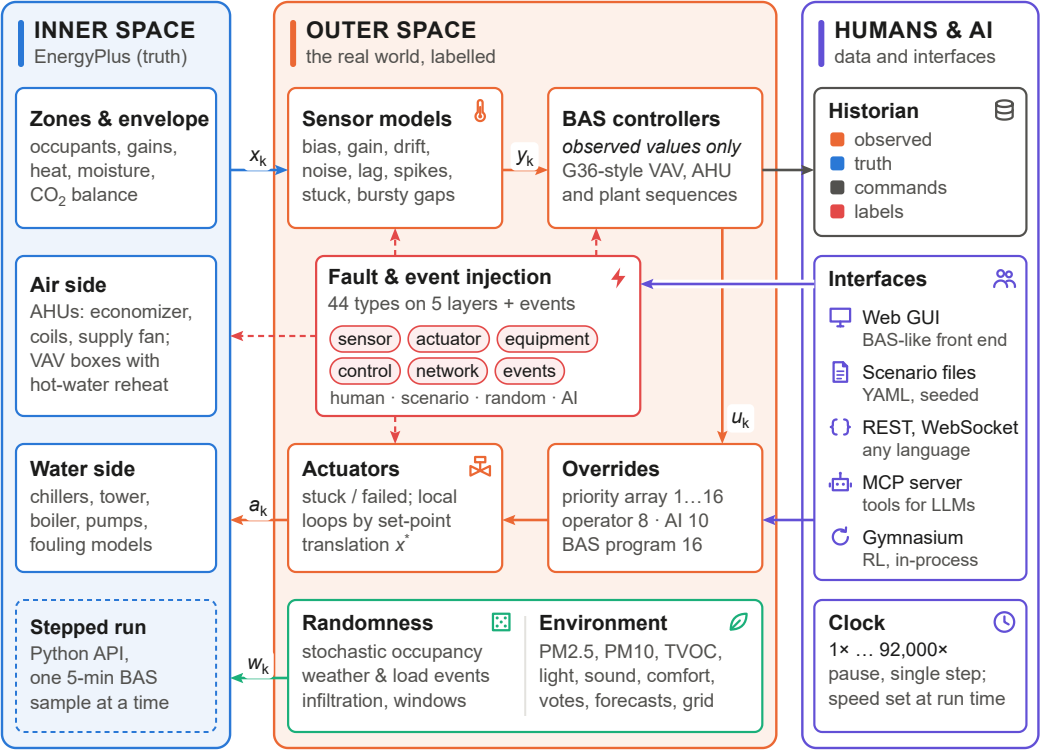

INNER SPACE
EnergyPlus (truth)
Zones & envelope
occupants, gains, heat, moisture, CO_2 balance
Air side
AHUs: economizer, coils, supply fan; VAV boxes with hot-water reheat
Water side
chillers, tower, boiler, pumps, fouling models
Stepped run
Python API, one 5-min BAS sample at a time
OUTER SPACE
the real world, labelled
Sensor models
bias, gain, drift, noise, lag, spikes, stuck, bursty gaps
BAS controllers
observed values only
G36-style VAV, AHU and plant sequences
Fault & event injection
44 types on 5 layers + events
sensor
actuator
equipment
control
network
events
human · scenario · random · AI
Actuators
stuck / failed; local loops by set-point translation x^*
Overrides
priority array 1…16
operator 8 · AI 10
BAS program 16
Randomness
stochastic occupancy
weather & load events
infiltration, windows
Environment
PM2.5, PM10, TVOC, light, sound, comfort, votes, forecasts, grid
x_k
y_k
u_k
a_k
w_k
HUMANS & AI
data and interfaces
Historian
observed
truth
commands
labels
Interfaces
Web GUI
BAS-like front end
Scenario files
YAML, seeded
REST, WebSocket
any language
MCP server
tools for LLMs
Gymnasium
RL, in-process
Clock
1× … 92,000×
pause, single step; speed set at run time